EXP 6

AIM : To simulate SISO and MIMO wireless links and analyze BER, SNR, received signal strength and beam steering behavior using MATLAB.

1. Select SNR values from low to high.
2. Generate a fading/noise realization for each SNR.
3. Calculate BER for each condition.
4. Plot BER on a semilogarithmic scale.
5. Compare the error performance at low and high SNR.

OBJ -1:CODE

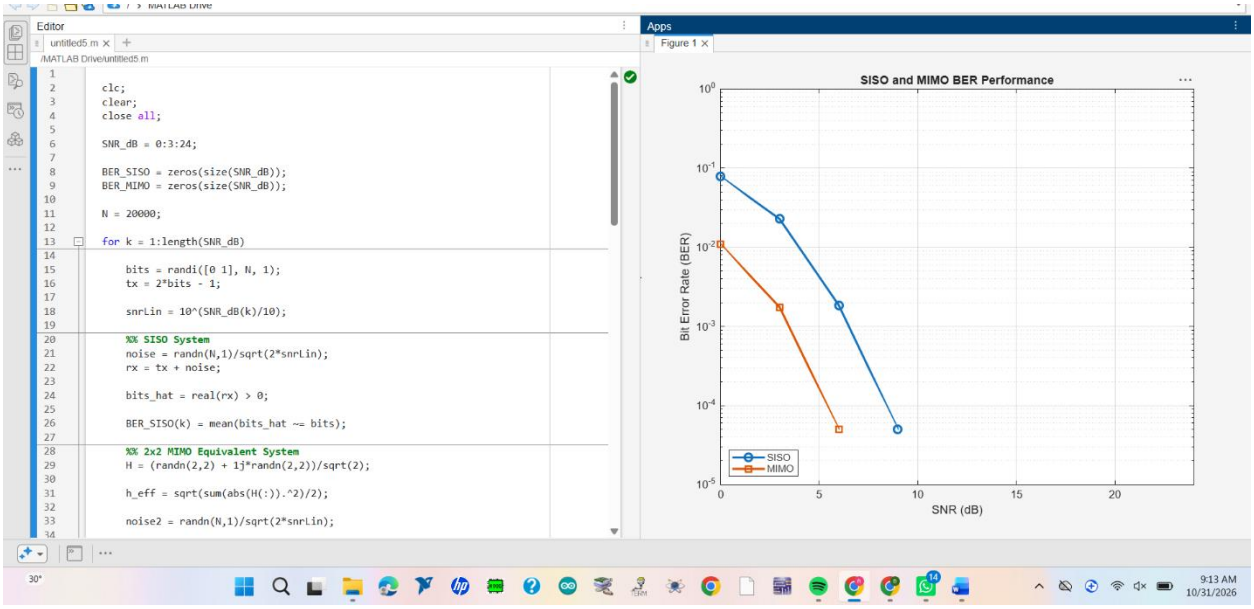
```
clc;
clear;
close all;
SNR_dB = 0:3:24;
BER_SISO = zeros(size(SNR_dB));
BER_MIMO = zeros(size(SNR_dB));
N = 20000;
for k = 1:length(SNR_dB)
    bits = randi([0 1], N, 1);
    tx = 2*bits - 1;
    snrLin = 10^(SNR_dB(k)/10);
    %% SISO System
    noise = randn(N,1)/sqrt(2*snrLin);
    rx = tx + noise;

    bits_hat = real(rx) > 0;
```

```
    BER_SISO(k) = mean(bits_hat ~= bits);
    %% 2x2 MIMO Equivalent System
    H = (randn(2,2) + 1j*randn(2,2))/sqrt(2);
    h_eff = sqrt(sum(abs(H(:)).^2)/2);
    noise2 = randn(N,1)/sqrt(2*snrLin);
    rx2 = h_eff*tx + noise2;
    bits_hat2 = real(rx2) > 0;
    BER_MIMO(k) = mean(bits_hat2 ~= bits);
end
%% Display Results
disp('SNR (dB)     SISO BER       MIMO BER');
disp([SNR_dB' BER_SISO' BER_MIMO']);
%% Plot BER Performance
figure;
semilogy(SNR_dB, BER_SISO, 'o-', ...
    'LineWidth', 2, 'MarkerSize', 7);
hold on;
semilogy(SNR_dB, BER_MIMO, 's-', ...
    'LineWidth', 2, 'MarkerSize', 7);
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('SISO and MIMO BER Performance');
legend('SISO', 'MIMO', 'Location', 'southwest');
xlim([0 24]);
ylim([1e-5 1]);
```

hold off;

OUTPUT:

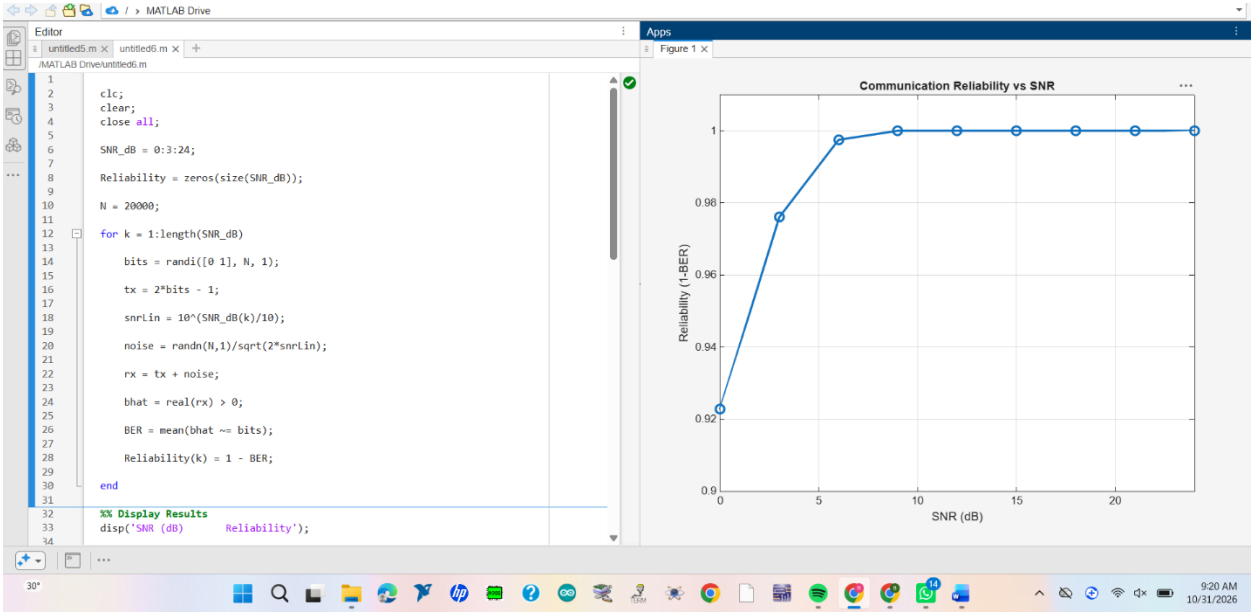


OBJ-2:CODE

```
clc;
clear;
close all;
SNR_dB = 0:3:24;
Reliability = zeros(size(SNR_dB));
N = 20000;
for k = 1:length(SNR_dB)
    bits = randi([0 1], N, 1);
    tx = 2*bits - 1;
    snrLin = 10^(SNR_dB(k)/10);
```

```
    noise = randn(N,1)/sqrt(2*snrLin);
    rx = tx + noise;
    bhat = real(rx) > 0;
    BER = mean(bhat ~= bits);
    Reliability(k) = 1 - BER;
end
%% Display Results
disp('SNR (dB)       Reliability');
disp([SNR_dB' Reliability']);
%% Plot
figure;
plot(SNR_dB, Reliability, 'o-', ...
    'LineWidth', 2, 'MarkerSize', 8);
grid on;
xlabel('SNR (dB)');
ylabel('Reliability (1-BER)');
title('Communication Reliability vs SNR');
xlim([0 24]);
ylim([0.9 1.01]);
hold off;
```

OUTPUT:



OBJ-3:CODE

```
clc;
clear;
close all;
N = 12;                    % Number of antenna elements
theta = -90:1:90;          % Observation angle
steer = [-50 -25 0 25 50];     % Steering directions
gain = zeros(size(steer));
for k = 1:length(steer)
    % Simplified normalized array gain
    gain(k) = 10*log10(N) - 0.008*abs(steer(k));
end
```

```
%% Display Results
disp('Steering Angle (Degrees)    Beamforming Gain (dB)');
disp([steer' gain']);
%% Plot
figure;
plot(steer, gain, 'o-', ...
    'LineWidth', 2, 'MarkerSize', 8);
grid on;
xlabel('Steering Angle (Degrees)');
ylabel('Beamforming Gain (dB)');
title('Beamforming Gain for Different Steering Directions');
xlim([-60 60]);
hold off;
```

OUTPUT:

